

Multiplying and dividing decimals

A Cambridge insert: count the decimal places for a product, and shift both numbers for a quotient.

LESSON 78–79

2 × 40 MIN

MATEMATIKA 6, QO‘SHIMCHA MAVZU

S7 4.3–4.4

LESSON CLOCK

25'

25'

25'

3

2

Multiplying	25 min
Dividing by a whole number	25 min
Dividing by a decimal	25 min
Checking	3 min
Homework	2 min

LEARNING OBJECTIVES

- Multiply decimals by counting the decimal places.
- Divide a decimal by a whole number.
- Divide by a decimal by shifting both numbers.
- Check every answer against an estimate.

TEXTBOOK

National	pp. 217–222
Cambridge	Stage 7, pp. 48–54
Workbook	Exercise 4.3

01

Explanation

Multiplying

Multiply as if there were no points, then put in as many decimal places as the two numbers had between them.

PRODUCT	WHOLE -NUMBER VERSION	PLACES	ANSWER
$0.3 \cdot 0.4$	$3 \cdot 4 = 12$	2	0.12
$1.2 \cdot 5$	$12 \cdot 5 = 60$	1	6.0
$2.5 \cdot 0.4$	$25 \cdot 4 = 100$	2	1.00
$6.9 \cdot 5.2$	$69 \cdot 52 = 3588$	2	35.88

$0.3 \cdot 0.4$ IS 0.12 , NOT 1.2

Two decimal places go in because the two numbers had one each. Multiplying two numbers below 1 gives an answer smaller than either of them.

Dividing by a whole number

Divide as usual and keep the decimal point above where it was.

QUOTIENT	WORKING	ANSWER
$7.2 \div 4$	$72 \div 4 = 18$	1.8
$0.35 \div 5$	$35 \div 5 = 7$	0.07
$9 \div 4$	$9.00 \div 4$	2.25
$1.2 \div 8$	$1.200 \div 8$	0.15

ZEROS MAY BE ADDED AFTER THE POINT

$9 \div 4$ does not stop at 2 remainder 1: writing 9.00 lets the division finish exactly, at 2.25.

Dividing by a decimal

Shift the point in **both** numbers until the divisor is a whole number. The quotient is unchanged, because both have been multiplied by the same power of ten.

QUOTIENT	SHIFTED	ANSWER
$6 \div 0.2$	$60 \div 2$	30
$4.8 \div 0.6$	$48 \div 6$	8
$1.44 \div 0.12$	$144 \div 12$	12
$0.9 \div 0.05$	$90 \div 5$	18

SHIFT BOTH, BY THE SAME AMOUNT

$6 \div 0.2$ becomes $60 \div 2$, not $6 \div 2$. Shifting only the divisor changes the answer by a factor of ten.

Checking

CALCULATION	ESTIMATE	EXACT
$6.9 \cdot 5.2$	$7 \cdot 5 = 35$	35.88
$0.3 \cdot 0.4$	less than 0.4	0.12
$6 \div 0.2$	much more than 6	30
$7.2 \div 4$	about 2	1.8

TWO QUESTIONS BEFORE EVERY ANSWER

Should it be bigger or smaller than what I started with, and roughly how big? A decimal point in the wrong place fails both tests immediately.

02 Worked examples

EXAMPLE 1 Compute $6.9 \cdot 5.2$.

- 1 Ignore the points: $69 \cdot 52 = 3\,588$.
- 2 One place plus one place makes two.
- 3 $= 35.88$
Estimate $7 \cdot 5 = 35$ ✓

ANSWER

35.88

EXAMPLE 2 Compute $4.8 \div 0.6$.

- ① Shift both one place: $48 \div 6$.

- ② $= 8$

- ③ Sense check: the divisor is under 1, so the answer beats 4.8 ✓

ANSWER

8

EXAMPLE 3 Compute $9 \div 4$.

- ① Write 9 as 9.00.

- ② $9.00 \div 4$

- ③ $= 2.25$

ANSWER

2.25

03 Interactive model

Ask for 0.3×0.4 before any rule; the popular answer of 1.2 is worth writing on the board and then testing against “less than a half of 0.4”.

Places and shifts

$0.3 \cdot 0.4$ equals:

1.2

0.12

0.012

12

$1.2 \cdot 5$ equals:

0.6

6

60

6.5

$6.9 \cdot 5.2$ equals:

35.88

3.588

358.8

34.88

$7.2 \div 4$ equals:

1.8

18

0.18

2.8

$9 \div 4$ equals:

2.2

2.25

2.5

2 r 1

$6 \div 0.2$ equals:

3

12

30

1.2

$1.44 \div 0.12$ equals:

1.2

12

120

0.12

Multiplying two numbers below 1 gives:

a larger answer

a smaller answer than both

the same

a whole number

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To multiply	Ko'paytirish	Умножать
To divide	Bo'lish	Делить
Decimal place	O'nli xona	Десятичный знак
Product	Ko'paytma	Произведение
Quotient	Bo'linma	Частное
To shift	Siljitmoq	Сдвигать
Whole-number divisor	Butun bo'luvchi	Целый делитель
Estimate	Baholash	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The number of decimal places in a product is:

the larger of the two

the sum of the two

always two

the smaller

$2.5 \cdot 0.4$ equals:

1

10

0.1

100

$0.35 \div 5$ equals:

0.7

0.07

7

0.007

To divide by 0.6 you first:

shift the dividend only

shift both one place

shift the divisor only

do nothing

$0.9 \div 0.05$ equals:

1.8

18

180

0.18

The best first check on a decimal answer is:

the digits

the size and the point

the sign

the units

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $0.3 \cdot 0.4$

ANSWER 0.12

2. $1.2 \cdot 5$

ANSWER 6

3. $2.5 \cdot 0.4$

ANSWER 1

4. $7.2 \div 4$

ANSWER 1.8

5. $0.35 \div 5$

ANSWER 0.07

6. $6 \div 0.2$

ANSWER 30

7. $4.8 \div 0.6$

ANSWER 8 **Medium · the standard question**

7 PROBLEMS

1. $6.9 \cdot 5.2$

ANSWER 35.88

2. $9 \div 4$

ANSWER 2.25

3. $1.2 \div 8$

ANSWER 0.15

4. $1.44 \div 0.12$

ANSWER 12

5. $0.9 \div 0.05$

ANSWER 18

6. $0.25 \cdot 0.8$

ANSWER 0.2

7. $3.6 \div 0.4$

ANSWER 9

Hard · stretch and reason

7 PROBLEMS

1. $0.125 \cdot 0.8$

ANSWER 0.1

2. $2.4 \cdot 1.5$

ANSWER 3.6

3. $7 \div 0.25$

ANSWER 28

4. $0.084 \div 0.7$

ANSWER 0.12

5. A pen costs 3.5 thousand sum: 12 pens

ANSWER 42 thousand

6. 1.6 kg shared into 0.2 kg bags

ANSWER 8 bags

7. Why does $6 \div 0.2$ give 30?

ANSWER Thirty two-tenths make six

07 Homework

Homework — 5 tasks

Estimate first, calculate second, and compare the two.

1. Compute $0.6 \cdot 0.7$, $3.4 \cdot 2.5$ and $1.25 \cdot 0.4$.
2. Compute $8.4 \div 6$ and $11 \div 8$.
3. Compute $9 \div 0.3$ and $2.4 \div 0.08$.
4. Estimate each answer above before finding it exactly.
5. A rope 7.2 m long is cut into 0.6 m pieces. How many pieces?